

Singular Value Decomposition (SVD)

$$\begin{pmatrix} 0 & X \\ X^T & 0 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = s \begin{pmatrix} u \\ v \end{pmatrix}$$

Singular Valued Decomposition, SVD, general linear maps

Singular Value Decomposition

- One of the most important results in linear algebra is that any matrix X can be decomposed as

$$X = USV^T$$

- If X is an $m \times p$ matrix then U is an $m \times m$ orthogonal matrix and V is a $p \times p$ orthogonal matrix i.e. $UU^T = U^T U = I$, $VV^T = V^T V = I$
- And S is an $m \times p$ matrix with non-zero diagonal elements $S_{ii} = s_i$ known as the **singular values**
- Where did that come from?

Eigenvectors

- Note that as $Xv = su$ and $X^T u = sv$ then

$$X(-v) = (-s)u \quad X^T u = (-s)(-v)$$

if $\begin{pmatrix} u \\ v \end{pmatrix}$ is an eigenvector of B with eigenvalue s then so is $\begin{pmatrix} u \\ -v \end{pmatrix}$ with eigenvalue $-s$

- If $m < p$ then $X^T X$ is not full rank so some eigenvalues are zero
- As a consequence $p - m$ vectors exist such that $Xv = 0$
- The eigenvalues and eigenvectors are

$$m \times \left(s_i, \begin{pmatrix} u_i \\ v_i \end{pmatrix} \right) \quad m \times \left(-s_i, \begin{pmatrix} u_i \\ -v_i \end{pmatrix} \right) \quad (p - m) \times \left(0, \begin{pmatrix} 0 \\ v_k \end{pmatrix} \right)$$

Normalisation Subtlety

$$\begin{pmatrix} 0 & X \\ X^T & 0 \end{pmatrix} = \begin{pmatrix} 0 & 2USV^T \\ 2VSU^T & 0 \end{pmatrix} = \begin{pmatrix} US & -US & 0 \\ VS & VS & 0 \end{pmatrix} = \begin{pmatrix} 0 & 2USV^T \\ 2VSU^T & 0 \end{pmatrix}$$

- Multiplying out we have $X = 2USV^T$ $X^T = 2VS^T U^T$
- Now the vectors u_i and v_i form orthogonal sets as they satisfy $X^T X v = s^2 v$ $XX^T u = s^2 u$
- But they are not normalised (since $\begin{pmatrix} u_i \\ v_i \end{pmatrix}$ is normalised). If we define $\tilde{U} = \sqrt{2}U$ and $\tilde{V} = \sqrt{2}V$ we find $X = \tilde{U}S\tilde{V}^T$ $X^T = \tilde{V}S^T\tilde{U}^T$

- Singular Value Decomposition
- General Linear Mappings
- Linear Regression Revisited

$$\begin{pmatrix} 0 & X \\ X^T & 0 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = s \begin{pmatrix} u \\ v \end{pmatrix}$$

Eigen Decomposition

- Consider an arbitrary $m \times p$ matrix X , and construct the $(m + p) \times (m + p)$ symmetric matrix, B ,

$$\begin{pmatrix} 0 & X \\ X^T & 0 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = s \begin{pmatrix} u \\ v \end{pmatrix}$$

$\begin{pmatrix} u \\ v \end{pmatrix}$ is an eigenvector of B with eigenvalue s

- We observe that

$$Xv = su \quad X^T u = sv$$

$$X^T X v = sX^T u = s^2 v \quad XX^T u = sXv = s^2 u$$

Matrix Decomposition

- Stacking the eigenvectors into a matrix

$$\begin{pmatrix} 0 & X \\ X^T & 0 \end{pmatrix} \begin{pmatrix} u & u & 0 \\ v & -v & v_0 \end{pmatrix} = \begin{pmatrix} u & u & 0 \\ v & -v & v_0 \end{pmatrix} \begin{pmatrix} s & 0 & 0 \\ 0 & -s & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

- Since the vectors $\begin{pmatrix} u_i \\ v_i \end{pmatrix}$ are eigenvectors of a symmetric matrix they form an orthogonal matrix if they are normalised.
- Multiply on the right by the transpose of the orthogonal matrix

$$\begin{pmatrix} 0 & X \\ X^T & 0 \end{pmatrix} = \begin{pmatrix} u & u & 0 \\ v & -v & v_0 \end{pmatrix} \begin{pmatrix} s & 0 & 0 \\ 0 & -s & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} u^T & v^T \\ u^T & -v^T \\ 0 & v_0^T \end{pmatrix}$$

SVD

- Any matrix, X , can be written as $X = USV^T$
 - U, V are orthogonal matrices
 - $S = \text{diag}(s_1, s_2, \dots, s_r)$ with $r = \min(m, p)$
- s_i can always be chosen to be non-negative and are known as **singular values**
- If X is symmetric and positive semi-definite (e.g. a covariance matrix) then $U = V$ and $S = \Lambda$ (the matrix of eigenvalues)—SVD is the same as eigen-decomposition (for a general symmetric matrix $s_i = |\lambda_i|$)
- Singular value decomposition applies to both square and non-square matrices—they describe general linear mappings

Spectrum of a Matrix

- The set of singular values (when ordered) is often referred to as the (singular value) spectrum of a matrix

- The largest singular value, s_{max} , provides a measure of the maximum change in the two norm of a vector

$$\mathbf{X}\mathbf{v}_{max} = \mathbf{U}\mathbf{S}\mathbf{V}^T\mathbf{v}_{max} = s_{max}\mathbf{u}_{max}$$

where $\mathbf{u}_{max}$ and $\mathbf{v}_{max}$ are the columns of the matrix $\mathbf{U}$ and $\mathbf{V}$ corresponding to s_{max} (note $\|\mathbf{v}_i\| = \|\mathbf{u}_i\| = 1$)

- $s_{max} = \|\mathbf{X}\|_2$ is known as the **spectral norm** or **2 norm** of a matrix satisfying

$$\|\mathbf{X}\mathbf{x}\|_2 \leq s_{max} \|\mathbf{x}\|_2$$

with equality when $\mathbf{x} = \mathbf{v}_{max}$

Economical Forms of SVD

- Often the rows or columns of the orthogonal matrices $\mathbf{U}$ and $\mathbf{V}$ that are not associated with a singular value are ignored

$$\begin{pmatrix} \mathbf{X} \\ \mathbf{0} \end{pmatrix} = \begin{pmatrix} \mathbf{U} \\ \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{S} \\ \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{V}^T \\ \mathbf{0} \end{pmatrix}$$

$$\begin{pmatrix} \mathbf{X} \\ \mathbf{0} \end{pmatrix} = \begin{pmatrix} \mathbf{U} \\ \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{S} \\ \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{V}^T \\ \mathbf{0} \end{pmatrix}$$

- In numpy (which returns a vector $\mathbf{s}$ and $\mathbf{V}^T$)

```
U, s, Vt = np.linalg.svd(X) # full: U is m x m
# economical: U is m x r
U, s, Vt = np.linalg.svd(X, full_matrices=False)
```

General Matrix

- Recall that we can compute the SVD for any matrix, $\mathbf{X}$

- As matrices describe the most general linear mapping

$$\mathbf{v} \rightarrow \mathcal{T}[\mathbf{v}] = \mathbf{X}\mathbf{v}$$

- We can use SVD to understand any linear mapping
- Thus any linear mapping can be seen as a rotation followed by a squashing or expansion independently in each coordinate followed by another rotation

Determinants

- The determinant, $|\mathbf{M}|$ of a matrix $\mathbf{M}$ is defined for square matrices

- It describes the change in volume under the mapping

- Now for any two matrices $|\mathbf{AB}| = |\mathbf{A}||\mathbf{B}|$

- Thus $|\mathbf{M}| = |\mathbf{U}||\mathbf{S}||\mathbf{V}^T|$

- For an orthogonal matrix $|\mathbf{U}| = \pm 1$ since $\mathbf{U}\mathbf{U}^T = \mathbf{I} \Rightarrow |\mathbf{U}\mathbf{U}^T| = |\mathbf{I}| \Rightarrow |\mathbf{U}||\mathbf{U}^T| = 1$ or $|\mathbf{U}|^2 = 1$

- Thus

$$|\mathbf{M}| = \pm |\mathbf{S}| = \pm \prod_i s_i$$

Finding SVD

- Most libraries will compute the SVD for you

- In principle we can choose the smaller of the two matrices $\mathbf{X}\mathbf{X}^T$ and $\mathbf{X}^T\mathbf{X}$ and compute its eigenvalues (libraries avoid this: it squares the condition number)

- The singular values are the square root of the eigenvalues (notice that $\mathbf{X}\mathbf{X}^T$ and $\mathbf{X}^T\mathbf{X}$ are both positive semi-definite so the eigenvalues will be non-negative)

- We can then compute the $\mathbf{U}$ matrix or $\mathbf{V}$ matrix by multiplying through by $\mathbf{X}$ or $\mathbf{X}^T$ ($\mathbf{U} = \mathbf{X}\mathbf{V}\mathbf{S}^{-1}$ and $\mathbf{V} = \mathbf{X}^T\mathbf{U}\mathbf{S}^{-1}$)

- In practice to perform PCA most people subtract the mean from their data and then perform SVD

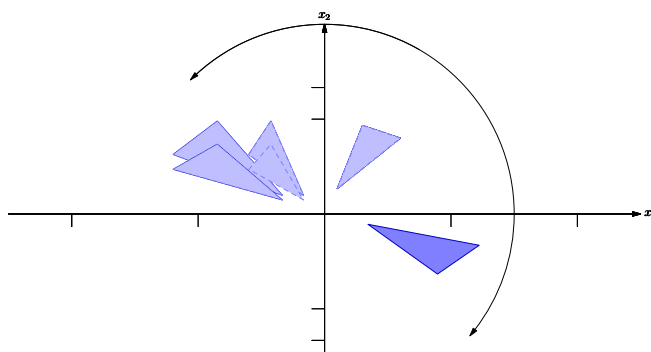
Outline

- Singular Value Decomposition
- General Linear Mappings
- Linear Regression Revisited

$$\begin{pmatrix} \mathbf{0} & \mathbf{X} \\ \mathbf{X}^T & \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{u} \\ \mathbf{v} \end{pmatrix} = \begin{pmatrix} \mathbf{u} \\ \mathbf{v} \end{pmatrix} \mathbf{s}$$

Matrices

$$\mathbf{M} = \begin{pmatrix} -0.45 & 1.9 \\ -0.77 & -0.025 \end{pmatrix} = \mathbf{U}\mathbf{S}\mathbf{V}^T = \begin{pmatrix} \cos(-175^\circ) & -\sin(-175^\circ) \\ \sin(-175^\circ) & \cos(-175^\circ) \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 0.75 \end{pmatrix} \begin{pmatrix} \cos(75^\circ) & -\sin(75^\circ) \\ \sin(75^\circ) & \cos(75^\circ) \end{pmatrix}$$



Non-Square Matrices

- When the matrices are non-square then the singular value matrix will either

★ Squash some directions to zero

★ Introduce new dimensions orthogonal to the vector

$$\begin{pmatrix} \mathbf{X} \\ \mathbf{0} \end{pmatrix} = \begin{pmatrix} \mathbf{U} \\ \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{S} \\ \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{V}^T \\ \mathbf{0} \end{pmatrix}$$

- The rank of an arbitrary matrix is the number of non-zero singular values (also number of linearly independent rows or columns)

Duality Revisited

- If $\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^T$ then

$$\begin{aligned}\mathbf{C} &= \mathbf{X}\mathbf{X}^T & \mathbf{D} &= \mathbf{X}^T\mathbf{X} \\ &= \mathbf{U}\mathbf{S}\mathbf{V}^T\mathbf{V}\mathbf{S}^T\mathbf{U}^T & &= \mathbf{V}\mathbf{S}^T\mathbf{U}^T\mathbf{U}\mathbf{S}\mathbf{V}^T \\ &= \mathbf{U}(\mathbf{S}\mathbf{S}^T)\mathbf{U}^T & &= \mathbf{V}(\mathbf{S}^T\mathbf{S})\mathbf{V}^T\end{aligned}$$

- If $\mathbf{X}$ is an $m \times p$ matrix then $\mathbf{S}\mathbf{S}^T$ is an $m \times m$ diagonal matrix with elements $S_{ii}^2 = s_i^2$
- $\mathbf{S}^T\mathbf{S}$ is a $p \times p$ matrix with elements $S_{ii}^2 = s_i^2$
- $\mathbf{U}$ and $\mathbf{V}$ are matrices of eigenvectors for $\mathbf{C}$ and $\mathbf{D}$
- The eigenvalues of $\mathbf{X}\mathbf{X}^T$ and $\mathbf{X}^T\mathbf{X}$ are $\lambda_i = S_{ii}^2 = s_i^2$

SS^T and S^TS

$$\mathbf{S} = \begin{pmatrix} s_1 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & s_2 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & s_m & 0 & 0 \dots & 0 \end{pmatrix}$$

$$\mathbf{S}^T\mathbf{S} = \begin{pmatrix} s_1^2 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & s_2^2 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & s_m^2 & 0 & 0 \dots & 0 \\ 0 & 0 & \dots & 0 & 0 & 0 \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & 0 & 0 \dots & 0 \end{pmatrix}$$

$$\mathbf{S}\mathbf{S}^T = \begin{pmatrix} s_1^2 & 0 & \dots & 0 \\ 0 & s_2^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & s_m^2 \end{pmatrix}$$

Having A Go

- It's really easy to verify this in Python

```
import numpy as np
X = np.random.rand(3, 2)
U, s, Vt = np.linalg.svd(X) # s: vector, Vt: V transposed
S = np.zeros_like(X)        # rebuild the 3 x 2 matrix S
np.fill_diagonal(S, s)
U @ S @ Vt                  # should equal X
U[:, 0] @ U[:, 1]           # columns of U are orthogonal
U.T @ U, U @ U.T            # both the identity
L, Va = np.linalg.eigh(X @ X.T) # eigenvalues, ascending
S @ S.T                      # the same values: s squared
```

- Test yourself!

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Linear Regression

- Given a set of data $\mathcal{D} = \{(\mathbf{x}_k, y_k) | k = 1, 2, \dots, m\}$
- In linear regression we try to fit a linear model

$$f(\mathbf{x}|\mathbf{w}) = \mathbf{x}^T\mathbf{w}$$

- Which we fit by minimising the squared error loss

$$L(\mathbf{w}) = \sum_{k=1}^m (f(\mathbf{x}_k|\mathbf{w}) - y_k)^2$$

Using SVD

- Using $\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^T$ then

$$\begin{aligned}\mathbf{X}^+ &= (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T \\ &= (\mathbf{V}\mathbf{S}^T\mathbf{S}\mathbf{V}^T)^{-1}\mathbf{V}\mathbf{S}^T\mathbf{U}^T \\ &= \mathbf{V}(\mathbf{S}^T\mathbf{S})^{-1}\mathbf{V}^T\mathbf{V}\mathbf{S}^T\mathbf{U}^T \\ &= \mathbf{V}(\mathbf{S}^T\mathbf{S})^{-1}\mathbf{S}^T\mathbf{U}^T = \mathbf{V}\mathbf{S}^+\mathbf{U}^T\end{aligned}$$

- If $m > p$

$$\mathbf{X}^T = \begin{pmatrix} | & | & | & | & | & | & | & | \\ s_1 & 0 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & s_2 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & 0 & s_3 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & s_p & 0 & 0 \dots & 0 \end{pmatrix}, \mathbf{S}^T = \begin{pmatrix} s_1 & 0 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & s_2 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & 0 & s_3 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & s_p & 0 & 0 \dots & 0 \end{pmatrix}$$

Matrix Form

- In matrix form we write $L(\mathbf{w}) = \|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2$

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1^T \\ \mathbf{x}_2^T \\ \vdots \\ \mathbf{x}_m^T \end{pmatrix} \quad \mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{pmatrix}$$

- Then $\nabla L(\mathbf{w}^*) = 0$ implies

$$\mathbf{w}^* = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} = \mathbf{X}^+\mathbf{y}$$

- This is known as the pseudo-inverse

Pseudo-Inverse of S

$$\mathbf{S}^T\mathbf{S} = \begin{pmatrix} s_1^2 & 0 & \dots & 0 \\ 0 & s_2^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & s_p^2 \end{pmatrix}, (\mathbf{S}^T\mathbf{S})^{-1} = \begin{pmatrix} s_1^{-2} & 0 & \dots & 0 \\ 0 & s_2^{-2} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & s_p^{-2} \end{pmatrix}$$

$$\mathbf{S}^+ = (\mathbf{S}^T\mathbf{S})^{-1}\mathbf{S}^T = \begin{pmatrix} s_1^{-1} & 0 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & s_2^{-1} & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & 0 & s_3^{-1} & \dots & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & s_p^{-1} & 0 & 0 \dots & 0 \end{pmatrix}$$

Moore–Penrose Pseudo-Inverse

- $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ needs every $s_i > 0$: it fails if $m < p$ or the columns of $\mathbf{X}$ are dependent
- The **Moore–Penrose** pseudo-inverse, $\mathbf{X}^+ = \mathbf{V} \mathbf{S}^+ \mathbf{U}^\top$, just skips the zero singular values: $S_{ii}^+ = 1/s_i$ if $s_i \neq 0$, and 0 otherwise
- **Over-constrained** ($m > p$, all $s_i > 0$): nothing changes, $\mathbf{w} = \mathbf{X}^+ \mathbf{y}$ minimises $\|\mathbf{X} \mathbf{w} - \mathbf{y}\|^2$
- **Under-constrained** ($m < p$): infinitely many $\mathbf{w}$ fit the data, and $\mathbf{X}^+ \mathbf{y}$ picks the one with the smallest $\|\mathbf{w}\|$
 - ★ If $s_i = 0$ then $\mathbf{X} \mathbf{v}_i = \mathbf{0}$: the data say nothing about $\mathbf{w}$ along $\mathbf{v}_i$, so that component is set to zero
- In numpy: `np.linalg.pinv` or `np.linalg.lstsq`

Regularisation

- Consider linear regression with a regulariser

$$\begin{aligned} \mathcal{L}(\mathbf{w}) &= \|\mathbf{X} \mathbf{w} - \mathbf{y}\|^2 + \eta \|\mathbf{w}\|^2 \\ &= \mathbf{w}^\top (\mathbf{X}^\top \mathbf{X} + \eta \mathbf{I}) \mathbf{w} - 2 \mathbf{w}^\top \mathbf{X}^\top \mathbf{y} + \mathbf{y}^\top \mathbf{y} \end{aligned}$$

- Thus

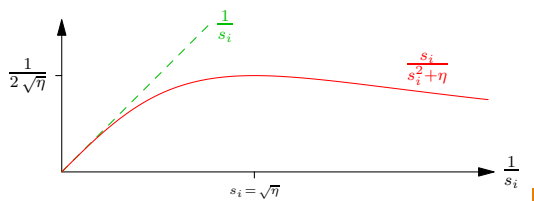
$$\nabla \mathcal{L}(\mathbf{w}) = 2(\mathbf{X}^\top \mathbf{X} + \eta \mathbf{I}) \mathbf{w} - 2 \mathbf{X}^\top \mathbf{y}$$

- and $\nabla \mathcal{L}(\mathbf{w}^*) = 0$ gives

$$\mathbf{w}^* = (\mathbf{X}^\top \mathbf{X} + \eta \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$$

Effect of Regularisation

- Without regularisation if $s_i = 0$ the problem would be ill-posed (even $\mathbf{S}^+$ does not exist since s_i^{-1} would be ill defined) and if s_i is small then $\mathbf{S}^+$ is ill-conditioned
- Using $\hat{\mathbf{S}}^+ = (\mathbf{S}^\top \mathbf{S} + \eta \mathbf{I})^{-1} \mathbf{S}^\top$ instead of $\mathbf{S}^+$ then



- Regularisation makes the machine much more stable (reduces the variance)

III-Conditioned Data Matrix

- Recall that

$$\mathbf{w}^* = \mathbf{X}^+ \mathbf{y} = \mathbf{V} \mathbf{S}^+ \mathbf{U}^\top \mathbf{y}$$

- If any of the singular values of $\mathbf{X}$ are small then $\mathbf{S}^+$ will magnify components in that direction
- Any errors in the target $\mathbf{y}$ will be magnified
- This leads to poor weights

Regularisation Continued

- Using $\mathbf{X} = \mathbf{U} \mathbf{S} \mathbf{V}^\top$

$$\begin{aligned} \mathbf{w}^* &= (\mathbf{X}^\top \mathbf{X} + \eta \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y} \\ &= \mathbf{V} (\mathbf{S}^\top \mathbf{S} + \eta \mathbf{I})^{-1} \mathbf{S}^\top \mathbf{U}^\top \mathbf{y} \end{aligned}$$

- where

$$(\mathbf{S}^\top \mathbf{S} + \eta \mathbf{I})^{-1} \mathbf{S}^\top = \begin{pmatrix} \frac{s_1}{s_1^2 + \eta} & 0 & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & \frac{s_2}{s_2^2 + \eta} & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & 0 & \frac{s_3}{s_3^2 + \eta} & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \frac{s_p}{s_p^2 + \eta} & 0 & 0 & \cdots & 0 \end{pmatrix}$$

Summary

- Any matrix can be decomposed as $\mathbf{X} = \mathbf{U} \mathbf{S} \mathbf{V}^\top$ where
 - ★ $\mathbf{U}$ and $\mathbf{V}$ are orthogonal (rotation matrices)
 - ★ $\mathbf{S} = \text{diag}(s_1, \dots, s_r)$ is a diagonal matrix of non-negative singular values
- This describes the most general linear transform
- The transform exploits the duality between $\mathbf{X} \mathbf{X}^\top$ and $\mathbf{X}^\top \mathbf{X}$
- In linear regression the pseudo-inverse involves the reciprocal of the singular values, which can lead to poor generalisation
- Regularisation improves the conditioning of the “inverse” matrix